

Implicit Integration for Density of States

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1 Quantum formalism

1.1 Periodic infinite crystal

Let $\mathcal{R}$ be a discrete periodic lattice of $\mathbb{R}^d$ (for $d \in \{1, \dots, 3\}$)

$$\mathcal{R} = \left\{ \sum_{i=1}^d n_i \mathbf{a}_i = A\mathbf{n}, n_i \in \mathbb{Z} \right\}, \quad (1)$$

where $A = (\mathbf{a}_1 \dots \mathbf{a}_d) \in \mathbb{R}^{d \times d}$ is non-degenerate.

We denote Γ its unit cell (the Wigner-Seitz cell centered around $\mathbf{0}$).

$$\Gamma = \{ \mathbf{r} \in \mathbb{R}^d, \|\mathbf{r}\|_2 \leq \|\mathbf{r} - \mathbf{R}\|_2, \forall \mathbf{R} \in \mathcal{R} \setminus \{\mathbf{0}\} \}. \quad (2)$$

Now let's define $\mathcal{R}^*$ the reciprocal lattice associated to $\mathcal{R}$

$$\mathcal{R}^* = \left\{ \sum_{i=1}^d n_i \mathbf{b}_i = B\mathbf{n}, n_i \in \mathbb{Z} \right\}, \quad (3)$$

where $A^\top B = 2\pi I$.

We now define the Brillouin zone $\mathcal{B}$ as the Wigner-Seitz cell centered around $\mathbf{0}$ in the reciprocal lattice

$$\mathcal{B} = \{ \mathbf{k} \in \mathbb{R}^d, \|\mathbf{k}\|_2 \leq \|\mathbf{k} - \mathbf{G}\|_2, \forall \mathbf{G} \in \mathcal{R}^* \setminus \{\mathbf{0}\} \} \quad (4)$$

1.2 Supercell method

For $L \in \mathbb{N}$, we define the supercell as

$$\Gamma_L = L\Gamma, \quad (5)$$

corresponding to L replication of the unit cell along each lattice direction. Now we naturally define the truncated lattice $\mathcal{R}_L$ as

$$\mathcal{R}_L = \mathcal{R} \cap \Gamma_L. \quad (6)$$

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This structure induces a discretization of the Brillouin zone by exploiting the duality between $\mathcal{R}$ and $\mathcal{R}^*$. We define this discretized Brillouin zone $\mathcal{B}_L$ as

$$\mathcal{B}_L = \left\{ \sum_{i=1}^d \frac{n_i}{L} \mathbf{b}_i \mid n_i \in \left\{ -\left\lfloor \frac{L}{2} \right\rfloor, \dots, \left\lfloor \frac{L}{2} - 1 \right\rfloor \right\} \right\}. \quad (7)$$

1.3 Hamiltonian and bands

Now we consider a Bloch Hamiltonian defined as the truncated Bloch transform of given real space Hamiltonians $H_{\mathbf{R}}$

$$H : \left\{ \mathbf{k} \mapsto \sum_{\mathbf{R} \in \mathcal{R}_L} e^{i\mathbf{k} \cdot \mathbf{R}} H_{\mathbf{R}} \right. \quad \mathcal{B} \rightarrow \mathbb{C}^{M \times M} \quad (8)$$

This Hamiltonian is $\mathcal{R}^*$ -periodic and is a Hermitian matrix which verifies the time-independent Schrödinger equation at each $\mathbf{k}$ of $\mathcal{B}$

$$H(\mathbf{k})\psi_m(\mathbf{k}) = \varepsilon_m(\mathbf{k})\psi_m(\mathbf{k}), \quad (9)$$

where ψ_m and ε_m respectively are the m-th eigenvectors and eigenvalues of H .

1.4 Density of states

The Density of States (DOS) is defined as

$$D(E) = \frac{1}{|\mathcal{B}|} \sum_{m=1}^M \int_{\mathcal{B}} \delta(E - \varepsilon_m(\mathbf{k})) d\mathbf{k} \quad (10)$$

A change of variable (co-area formula) gives another expression

$$D(E) = \frac{1}{|\mathcal{B}|} \sum_{m=1}^M \int_{S_m(E)} \frac{1}{|\nabla \varepsilon_m(\mathbf{k})|} d\mathcal{H}^{d-1}(\mathbf{k}) \quad (11)$$

where $d\mathcal{H}^{d-1}(\mathbf{k})$ is the Hausdorff measure of dimension $d-1$ and the $S_m(E)$ are isosurfaces of the m-th eigenvalue

$$S_m(E) = \{\mathbf{k} \in \mathcal{B}, \varepsilon_m(\mathbf{k}) = E\}. \quad (12)$$

From Equation 23, we can define the DOS as a sum of integrals over implicitly defined surfaces, the $S_m(E)$. Therefore the method of implicit integration [Say15] can be meaningful if we manage to bound efficiently the different eigenvalues on intervals.

2 High-Order Quadrature Methods for Implicitly Defined Surfaces and Volumes in Hyperrectangles

The idea of this algorithm is to compute integrals of sufficiently smooth function f on implicitly defined surfaces (with $U \subset \mathbb{R}^d$ a given hyperrectangle).

$$\int_{\Gamma \cap D} f \quad (13)$$

where Γ is a surface defined implicitly $\Gamma = \{x \in \mathbb{R}^d \mid \phi(x) = 0\}$.

2.1 Quadrature on implicitly defined domains via dimension reduction

The idea is to convert Γ into the graph of an implicitly defined height function. To do so we define an d -dimensional hyperrectangle U

$$U = (x_1^L, x_1^U) \times \cdots \times (x_d^L, x_d^U) \quad (14)$$

and we denote by $(e_i)_{1 \leq i \leq d}$ the canonical basis of $\mathbb{R}^d$.

For a sufficiently small U and a smooth enough ϕ (at least $\mathcal{C}^1$), we have $|\partial_{x_i} \phi|$ which is bounded away from zero. Therefore we can apply the implicit function theorem to describe locally $\Gamma \cap U$ with a height function such that $x_i = h(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_d)$ on $[x_i^L, x_i^U]$.

$$\phi(x_1, \dots, x_{i-1}, h(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_d), x_{i+1}, \dots, x_d) = 0 \quad (15)$$

This is illustrated in Figure 1.

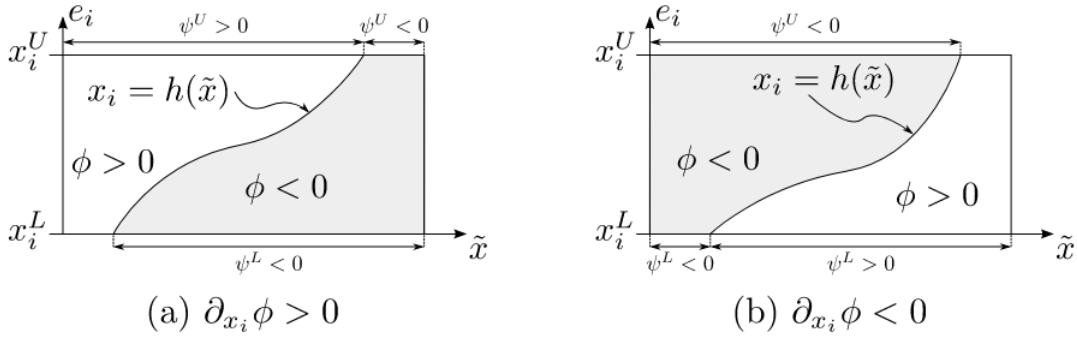


Figure 1: Cases for the definition of V_1 and V_2 (image from [Say15])

We define two functions ψ^L and ψ^U of $\tilde{x} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_d)$ which are the restrictions of ϕ to the lower and upper faces.

$$\psi^L(\tilde{x}) = \phi(\tilde{x} + x_i^L e_i) \quad (16)$$

$$\psi^U(\tilde{x}) = \phi(\tilde{x} + x_i^U e_i) \quad (17)$$

This allows to define 2 $(d - 1)$ -dimensional volumes V_1 and V_2

$$V_1 = \begin{cases} \{\tilde{x} : \psi^L(\tilde{x}) < 0\} & \text{if } \partial_{x_i}\phi > 0 \text{ in } U \\ \{\tilde{x} : \psi^L(\tilde{x}) > 0\} & \text{if } \partial_{x_i}\phi < 0 \text{ in } U \end{cases} \quad (18)$$

$$V_2 = \begin{cases} \{\tilde{x} : \psi^U(\tilde{x}) > 0\} & \text{if } \partial_{x_i}\phi > 0 \text{ in } U \\ \{\tilde{x} : \psi^U(\tilde{x}) < 0\} & \text{if } \partial_{x_i}\phi < 0 \text{ in } U \end{cases} \quad (19)$$

From (15) we have $dS = \sqrt{1 + |\nabla_{\tilde{x}}h(\tilde{x})|^2}d\tilde{x}$. Moreover by derivating (15),

$$\forall j \neq i, \partial_{x_j}h(\tilde{x}) = -\frac{\partial_j\phi}{\partial_i\phi} \Big|_{x=\tilde{x}+h(\tilde{x})e_i} \quad (20)$$

Thus

$$\sqrt{1 + |\nabla_{\tilde{x}}h(\tilde{x})|^2} = \frac{|\nabla\phi|}{|\partial_i\phi|} \Big|_{x=\tilde{x}+h(\tilde{x})e_i} \quad (21)$$

Using this equality yields

$$\int_{\Gamma \cap U} g dS = \int_{V_1 \cap V_2} g \frac{|\nabla\phi|}{|\partial_i\phi|} \Big|_{x=\tilde{x}+h(\tilde{x})e_i} d\tilde{x} \quad (22)$$

Therefore this algorithm convergence depends on the correct localization of the isosurface Γ , and this localization itself intrinsically depends on the quality of the bounds on ϕ and its partial derivatives.

2.2 Application to eh computation of the Density of States

Now we want to apply this algorithm to the Density of States (DOS). We remind that

$$D(E) = \frac{1}{|\mathcal{B}|} \sum_{m=1}^M \int_{S_m(E)} \frac{1}{|\nabla\varepsilon_m(\mathbf{k})|} d\mathcal{H}^{d-1}(\mathbf{k}), \quad (23)$$

which can be computed using the above-presented algorithm applied at each eigenvalue minus the desired energy level. We write

$$\phi_{m,E} = \varepsilon_m - E \quad (24)$$

In our case we will focus on the case of boxes centered around points $\mathbf{k}_0$, we will denote the box $\delta\mathbf{k}$ and the box centered around $\mathbf{k}_0$ as $D = \delta\mathbf{k} + \mathbf{k}_0$

3 Time-independent perturbation theory

We want to find approximations of the eigenvalues ε_m and eigenstates ψ_m of a Hamiltonian H . To do so we express H as a small perturbation of a reference Hamiltonian H_0 by a

small parameter λ . Then using a serie expansion in λ we can find approximations of H eigenvalues/eigenstates depending on H_0 eigenvalues $\varepsilon_{m,0}$ and eigenstates $\psi_{m,0}$. Let H , H_0 and δH be three Hamiltonian of $\mathbb{C}^{M \times M}$ such that

$$H(\lambda) = H_0 + \lambda \delta H \quad (25)$$

where λ is a small positive parameter.

We define for $m \in \llbracket 1, M \rrbracket$

$$H\psi_m = \varepsilon_m \psi_m \quad (26)$$

and

$$H_0\psi_{m,0} = \varepsilon_{m,0}\psi_{m,0} \quad (27)$$

Now considering a non-degenerate eigenvalues $\varepsilon_m(\lambda)$. We expand it in series as

$$\varepsilon_m(\lambda) = \sum_{i=0}^{+\infty} \lambda^i \varepsilon_m^{(i)} \quad (28)$$

Using the normalization of Hamiltonian eigenstates ($\|\psi\|_2 = 1$) and identifying each term in the development we have at second-order that

$$\varepsilon_m(\lambda) = \varepsilon_{m,0} + \lambda \langle \psi_{m,0} | W | \psi_{m,0} \rangle + \lambda^2 \sum_{n \neq m} \frac{\langle \psi_{n,0} | W | \psi_{m,0} \rangle^2}{\varepsilon_{m,0} - \varepsilon_{n,0}} + \mathcal{O}(\lambda^3) \quad (29)$$

Now considering a sufficiently smooth (at least continously differentiable) Hamiltonian $H_{\mathbf{k}}$ at point $\mathbf{k}$ of the Brillouin zone. We want its Taylor expansion around point $\mathbf{k}_0$, we write $\delta_{\mathbf{k}} = (\delta_{k_i})_{i \in \{1, \dots, d\}} = (k_i - k_{0,i})_{i \in \{1, \dots, d\}}$. Here the perturbation theory yields

$$\varepsilon_m(\mathbf{k}) = \varepsilon_m(\mathbf{k}_0) + \sum_{i \in \{1, \dots, d\}} \frac{\partial \varepsilon_m(\mathbf{k}_0)}{\partial k_i} \delta_{k_i} + \frac{1}{2} \sum_{i, j \in \{1, \dots, d\}} \frac{\partial^2 \varepsilon_m(\mathbf{k}_0)}{\partial k_i \partial k_j} \delta_{k_i} \delta_{k_j} + \mathcal{O}(\|\delta_{\mathbf{k}}\|^3) \quad (30)$$

where

$$\frac{\partial \varepsilon_m}{\partial k_i} = \langle \psi_m | \partial_i H | \psi_m \rangle \quad (31)$$

and

$$\frac{\partial^2 \varepsilon_m}{\partial k_i \partial k_j} = \langle \psi_m | \partial_i \partial_j H | \psi_m \rangle + 2 \sum_{n \neq m} \frac{\text{Re}(\langle \psi_m | \partial_i H | \psi_n \rangle \langle \psi_n | \partial_j H | \psi_m \rangle)}{\varepsilon_m - \varepsilon_n} \quad (32)$$

3.1 Guaranteed theoretical bound

Now from Equation 30 we can derive guaranteed bounds on the eigenvalues.

Introduire les notations avant pour H , ε_m, ψ_m , $\delta \mathbf{k}$ et D

Lemma 3.1. *Weyl's Inequality for Bloch eigenvalues*

Let $\mathbf{k}_0 \in \mathcal{B}$ and $H : \mathcal{B} \rightarrow \mathbb{C}^{M \times M}$ an analytic Bloch Hamiltonian. Let $D = \mathbf{k}_0 + \delta \mathbf{k} = \{\mathbf{k} \in \mathcal{B}, |k_i - k_{0,i}| \leq \delta k_i, i \in \{1, \dots, d\}\}$ the hyperrectangle centered around $\mathbf{k}_0$. Then for all $m = 1, \dots, M$, $\mathbf{k} \in D$,

$$|\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0)| \leq \sup_{\mathbf{q} \in D} \|H(\mathbf{q}) - H(\mathbf{k}_0)\|_{\text{op}} \quad (33)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm defined as

$$\|H\|_{\text{op}} = \sup_{\|\psi\|_2=1} \|H\psi\|_2. \quad (34)$$

Corollary 3.1 (First-order bound). *In addition, if ε_m is non-degenerate, then for all $\mathbf{k} \in D$,*

$$|\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0)| \leq \sup_{\mathbf{k} \in D} \sum_{j=1}^d \|\partial_j H(\mathbf{k})\|_{\text{op}} |\delta k_j| \leq \sup_{\mathbf{k} \in D} \sqrt{\sum_{j=1}^d \|\partial_j H(\mathbf{k})\|_{\text{op}}^2} \|\delta \mathbf{k}\|_2 \quad (35)$$

Proof. Let $\mathbf{k} \in D$, we suppose that each eigenvalue is non-degenerate in D .

Let $\gamma : t \mapsto (1-t)\mathbf{k}_0 + t\mathbf{k}$ a straight-line path between $\mathbf{k}$ and $\mathbf{k}_0$. We define $f : t \mapsto H(\gamma(t))$ which is analytic by composition. The chain rule yields

$$f'(t) = \sum_{j=1}^d \partial_j H(\gamma(t)) (k_j - k_{0,j}). \quad (36)$$

Hence

$$H(\mathbf{k}) - H(\mathbf{k}_0) = f(1) - f(0) = \int_0^1 f'(t) dt = \sum_{j=1}^d \int_0^1 \partial_j H(\gamma(t)) (k_j - k_{0,j}) dt. \quad (37)$$

Taking the operator norm and applying the triangular inequality gives

$$\|H(\mathbf{k}) - H(\mathbf{k}_0)\|_{\text{op}} \leq \sum_{j=1}^d \int_0^1 \|\partial_j H(\gamma(t))\|_{\text{op}} |k_j - k_{0,j}| dt \leq \sum_{j=1}^d \|\partial_j H(\gamma(t))\|_{\text{op}} |\delta k_j|. \quad (38)$$

Now by combining that $\forall t \in [0, 1], \gamma(t) \in D$ and the Lemma 3.1

$$|\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0)| \leq \|H(\mathbf{k}) - H(\mathbf{k}_0)\|_{\text{op}} \leq \sup_{\mathbf{q} \in D} \sum_{j=1}^d \|\partial_j H(\mathbf{q})\|_{\text{op}} |\delta k_j|. \quad (39)$$

Finally, Cauchy-Schwarz inequality gives the second bound

$$|\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0)| \leq \sup_{\mathbf{k} \in D} \sqrt{\sum_{j=1}^d \|\partial_j H(\mathbf{k})\|_{\text{op}}^2} \|\delta \mathbf{k}\|_2 \quad (40)$$

This concludes the proof. $\square$

Corollary 3.2 (Second-order bound). *Under the same non-degeneracy hypothesis on ε_m , we have for all $\mathbf{k} \in D$,*

$$|\varepsilon_m(\mathbf{k}) - \varepsilon_m(\mathbf{k}_0) - \sum_{i=1}^d \delta k_i \frac{\partial \varepsilon_m}{\partial k_i}(\mathbf{k}_0)| \leq \frac{1}{2} \|\delta \mathbf{k}\|_2^2 \sup_{\mathbf{k} \in D} \left(\|\nabla^2 H(\mathbf{k})\|_{\text{op}} + 2 \frac{\|\nabla H(\mathbf{k})\|_{\text{op}}^2}{\Delta_m} \right) \quad (41)$$

where $\Delta_m = \min_{\mathbf{k}_1, \mathbf{k}_2 \in D, n \neq m} |\varepsilon_m(k_1) - \varepsilon_n(k_2)|$ is the gap between the m -th eigenvalue and the closest eigenvalue of a different band.

Proof. The idea is to use the second term of the Taylor expansion of our eigenvalue. Let $\mathbf{k} \in D$, □

3.2 Numerical bounds

The theoretical bounds are valid, but we often don't know the exact supremum of our eigenvalues on a given box. To ensure our algorithm performs reliably, we need numerical estimates. One approach is to use interval arithmetic. Although the bounds from interval arithmetic aren't sharp, they are rigorous—especially since our Hamiltonians are represented as truncated Fourier series of known matrices. Our strategy, then, is to combine and compare several different eigenvalue bounds to find the most effective estimate.

3.2.1 Interval Arithmetic

We remind the Taylor expansion for functions over an interval

$$[f]([x]) = f(y) + \sum_{|\alpha|=1}^k \frac{D^\alpha f(y)}{\alpha!} ([x] - y)^\alpha + \sum_{|\alpha|=k+1} \frac{[D^{|\alpha|} f]([x])}{(\alpha)!} ([x] - y)^\alpha \quad (42)$$

where $\alpha = (\alpha_1, \dots, \alpha_d)$, $\alpha! = \prod_{i=1}^d \alpha_i!$, $D^\alpha f = \frac{\partial^\alpha f}{\partial k_1^{\alpha_1} \dots \partial k_d^{\alpha_d}}$ and $([x] - y)^\alpha = \prod_{i=1}^d ([x] - y)^{\alpha_i}$.

In our case up to first and second order we have

$$[\varepsilon_m]([\delta \mathbf{k}] + \mathbf{k}_0) = \varepsilon_m(\mathbf{k}_0) + \sum_{i=1}^d [\partial_i \varepsilon_m]([\delta \mathbf{k}] + \mathbf{k}_0) [\delta k_i] \quad (43)$$

$$[\varepsilon_m]([\delta \mathbf{k}] + \mathbf{k}_0) = \varepsilon_m(\mathbf{k}_0) + \sum_{i=1}^d \partial_i \varepsilon_m(\mathbf{k}_0) \cdot [\delta k_i] + \sum_{i,j=1}^d \frac{[\partial_i \partial_j \varepsilon_m]([\delta \mathbf{k}] + \mathbf{k}_0)}{2!} \cdot [\delta k_i] [\delta k_j] \quad (44)$$

3.2.2 Sampling bounds

The idea of sampling bounds is to get non-guaranteed bounds but which tends to the real bounds as the number of sampling points tends to infinity. The idea is to use the bounds obtained in Lemma 3.1 and to obtain an estimation of the sup of the desired quantities to directly use our bounds.

Cette partie là on fait pas du coup si j'ai bien compris

4 Bounds for degenerate eigenvalues

For degenerate eigenvalues, the Hamiltonian regularity does not imply the regularity of its eigenvalues. Thus the classical perturbation theory is not fitted to bound the eigenvalues after the first-order.

Let $\mathbf{k}_0 \in \mathcal{B}$ such that $\varepsilon_i, \varepsilon_j$ are degenerate eigenvalues. Let P be the projector on the eigenspace spanned by the corresponding eigenvectors ψ_i, ψ_j

$$P = |\psi_i\rangle \langle \psi_i| + |\psi_j\rangle \langle \psi_j| \quad (45)$$

Applying this projector to H yields an effective Hamiltonian H_{eff} which is a 2×2 Hermitian matrix.

$$H_{\text{eff}}(\mathbf{k}) = PH(\mathbf{k})P \quad (46)$$

Now using the Pauli decomposition for Hermitian matrices yields at $\mathbf{k}$ near $\mathbf{k}_0$ yields

$$H_{\text{eff}}(\mathbf{k}) = a_0(\mathbf{k})I + \mathbf{a}(\mathbf{k}) \cdot \boldsymbol{\sigma} \quad (47)$$

where

$$\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) \quad (48)$$

are the Pauli matrices.

Now for the sake of simplicity let $a_0(\mathbf{k}) = 0$ (as it just shifts both bands equally).

The eigenvalues of such a Hamiltonian are

$$\varepsilon_{\pm}(\mathbf{k}) = \pm \|\mathbf{a}(\mathbf{k})\|_2 \quad (49)$$

Consequently for general Hamiltonian the only way to have a crossing is such that $\mathbf{a}(\mathbf{k}) = \mathbf{0}$. Near the degeneracy (where the bands are still continuously differentiable) a first-order Taylor expansion yields

$$\mathbf{a}(\mathbf{k}) = \underbrace{\mathbf{a}(\mathbf{k}_0)}_{=0, \text{ since it is the crossing point}} + \nabla_{\mathbf{k}} \mathbf{a}(\mathbf{k}_0) \cdot (\mathbf{k} - \mathbf{k}_0) + \mathcal{O}(\|\mathbf{k} - \mathbf{k}_0\|^2) \quad (50)$$

Thus locally around $\mathbf{k}_0$,

$$\mathbf{a}(\mathbf{k}) \approx \sum_{i=1}^d \frac{\partial \mathbf{a}(\mathbf{k}_0)}{\partial k_i} \delta k_i \quad (51)$$

However in reality all lattices present symmetries that change this behaviour and crossings of dimension 1 and 2 can happen.

References

- [Say15] R. I. Saye, *High-Order Quadrature Methods for Implicitly Defined Surfaces and Volumes in Hyperrectangles*, SIAM Journal on Scientific Computing **37** (2015), no. 2, A993–A1019, Publisher: Society for Industrial and Applied Mathematics.